Shenghan **Gao**

OPTICS · PHOTONICS · COMPUTER SCIENCE

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Summary

Ph.D. candidate in Electrical and Computer Engineering with research depth in integrated photonics, coherent optical systems, and fiber-optic sensing, plus an M.S. Computer Science background in machine learning. Build and characterize photonic chips and optical testbeds using lasers, modulators, free-space/fiber optics, OSAs, SVAs, lock-in amplifiers, and photodetectors. Develop object-oriented Python automation for instrument control, repeatable parameter sweeps, batch data acquisition, and model-to-measurement analysis. Seeking research or R&D roles where photonics, experimental systems, and software-driven measurement intersect.

Education

University of Pittsburgh — *Pittsburgh, PA*

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING, DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING — *Dec. 2026 (expected)*

Georgia Institute of Technology — *Atlanta, GA*

M.S. IN COMPUTER SCIENCE (MACHINE LEARNING TRACK) — *May 2026 (Expected)*

University of Rochester — *Rochester, NY*

M.S. IN OPTICS, THE INSTITUTE OF OPTICS (THESIS TRACK) — *Aug. 2019*

B.S. IN OPTICAL ENGINEERING, THE INSTITUTE OF OPTICS — *May 2017*

B.S. IN APPLIED MATH, DEPARTMENT OF MATHEMATICS — *May 2017*

- Related coursework: silicon photonics simulation, signal processing, software engineering, deep learning, reinforcement learning, circuit design, lens design, coating design, photonic device design, and computational imaging.

Work Experience

Lumentum Operations LLC — *San Jose, CA*

RESEARCH INTERN | MENTOR: JOE WEN — *May 2023 - Aug 2023*

- Investigated FIR-filter-based equalization for communication signals affected by intersymbol interference, connecting filter design choices to recovered signal quality.
- Collected and post-processed measured signal data, using FFT-based frequency analysis to diagnose distortion and compare filter behavior across test cases.
- Implemented FIR filter design workflows in MATLAB and Python, tuning coefficients with optimization models to improve demodulated-signal restoration and bandwidth handling.

Sunny Optics — *Ningbo, China*

RESEARCH INTERN | MENTOR: CHAO XU — *May 2018 - Aug 2018*

- Designed automotive LiDAR lens systems in CodeV and Zemax, translating client optical specifications into simulated lens layouts and optimization targets.
- Collaborated with opto-mechanical engineers during lab integration, using test feedback to refine optical alignment and system functionality.
- Optimized mobile camera lens designs in Zemax through iterative simulation, balancing imaging performance with packaging constraints.

Research Experience

University of Pittsburgh — *Pittsburgh, PA*

RESEARCH ASSISTANT | ADVISOR: NATHAN YOUNGBLOOD — *Sep 2019 - present*

- Led experimental development of remote optical injection locking between a long-haul master laser and a local slave laser, enabling stable phase-coherent operation for interference-based signal processing, coherent detection, and optical-domain computing.
- Designed integrated photonic waveguides and components in KLayout, then validated fabricated chips through e-beam lithography workflows and optical characterization of coupling loss, interference contrast, and modulator response.
- Developed device- and system-level models in Ansys Lumerical, linking modulator efficiency, insertion loss, and coupling loss to locking range, SNR, and bandwidth; compared model predictions with measured system behavior.
- Built object-oriented Python automation with NI MAX and PyVISA/SCPI to coordinate lasers, SVAs, OSAs, and related instruments, creating reusable instrument and experiment classes for parameter sweeps, batch acquisition, and standardized analysis.
- Fabricated tapered fibers with embedded Fiber Bragg Gratings and designed fiber packaging for stress and temperature sensing; used COMSOL multiphysics to optimize taper geometry and thermo-mechanical response.
- Built a BOTDR setup for distributed temperature sensing and developed FBG/BOTDR calibration procedures, including temperature-wavelength mapping, sensitivity extraction, system-drift compensation, and regression-based analysis of uncertainty, noise, repeatability, and linearity.

University of Rochester *Rochester, NY*

Research Assistant | Advisor: Xi-Cheng Zhang *Sep 2017 - Aug 2019*

- Generated THz signals from liquid plasmas under varying temperature, polarity, and viscosity, quantifying how liquid properties influenced emission strength and spectral content.
- Used high-intensity ultrafast lasers to machine micron-scale apertures in syringe needles, controlling gas flow to produce acoustic tone spectra for gas-acoustic interaction studies.
- Implemented a digital cloaking prototype using commercially available lenslet arrays and MATLAB-based pixel remapping.
- Processed experimental data in time and frequency domains, using SNR, repeatability, sensitivity analysis, and regression fits to connect experimental parameters with sensing and THz performance.

Technical Skills

Photonics and optical hardware

- Integrated photonics, optical injection locking, fiber-optic sensing, FBG/BOTDR systems, microfabrication, e-beam lithography, etching, SEM, system/board design, and circuit-board layout.
- Optical testbeds with lasers, modulators, free-space and fiber optics, spectrometers/OSAs, SVAs, lock-in amplifiers, and photodetectors; trained in class 3B/4 laser safety.

Software, modeling, and analysis

- Python (NumPy, SciPy, Pandas, Matplotlib, PyVISA/SCPI, pytest), MATLAB, C++, Git/GitHub, instrument-control automation, batch acquisition, signal processing, and regression-based analysis.
- Ansys Lumerical, RSOFT, MEEP, RCWA, COMSOL, Zemax OpticStudio, CodeV, LightTools, LTspice, SolidWorks, NX, scikit-learn, PyTorch, TensorFlow, OpenCV.

Languages

Chinese (Native), English (Fluent), Japanese (Fluent)

Publications and Presentations

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., K. Lüdge, F. Da Ros, N. Youngblood, submitted to APL, under review, 2026

Distributed Coherent Optical Computing via Injection-Locked Photonic Networks

Gao, S., Invited PhD Candidate Talk, Pittsburgh Quantum Institute, Pittsburgh, 2025

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, Optical Waveguide and Laser Sensors II, SPIE, 2023

Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K (oral presentation)

Gao, S., P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, K. P. Chen, SPIE Defense + Commercial Sensing, Orlando, 2023

Preference of subpicosecond laser pulses for terahertz wave generation from liquids

Q. Jin, Y. E., **S. Gao**, X.-C. Zhang, *Advanced Photonics* 2(1), 015001, 2020

Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation

S. Gao, Q. Jin, Y. E., X.-C. Zhang, Infrared, Millimeter-Wave, and Terahertz Technologies VI, SPIE, 2019

Investigation of liquid lines as terahertz emitters under ultrashort optical excitation

Q. Jin, **S. Gao**, X.-C. Zhang, arXiv:1902.07150, 2019

Fabrication and testing of the smallest flute on syringe needles

S. Gao, Z. Jin, Y. E., X.-C. Zhang, Advanced Laser Processing and Manufacturing III, SPIE, 2019